DAQ Core

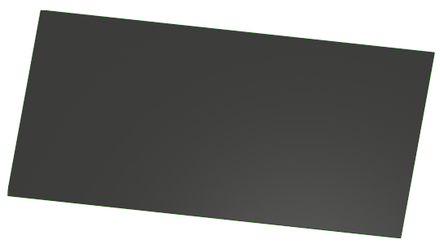
Variant: DRAFT

2026-11-06

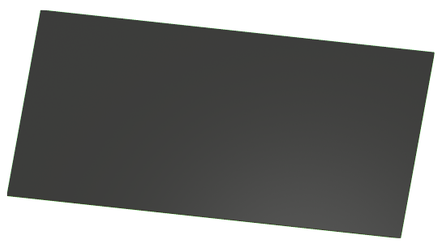
Rev + (Unreleased)

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11	Digital - In&Out	21	Power - Sequencing	31	.....
2	Block Diagram	12	Command - High Power	22	Revision History	32	.....
3	Project Architecture	13	Command - LED	23	.....	33	.....
4	MCU	14	Interface - RS232 & RS485	24	.....	34	.....
5	MCU - Programming	15	Interface - CAN	25	.....	35	.....
6	MCU - Expander	16	Interface - Other	26	.....	36	.....
7	MCU - IOT	17	Sensor - Environment	27	.....	37	.....
8	Analog - Input 3V3	18	Sensor - IMU	28	.....	38	.....
9	Analog - Input 5V & 10V	19	Memory	29	.....	39	.....
10	Analog - Output	20	Power	30	.....	40	.....

TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 06-Nov-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational design notes.

DESIGN NOTE:
Example text for debug notes.

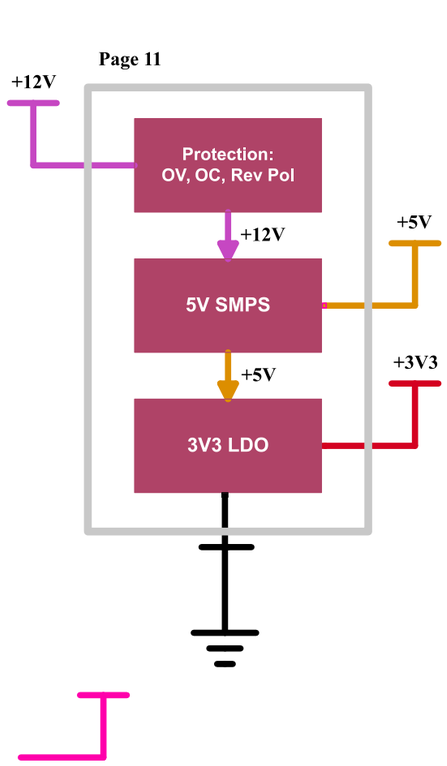
DESIGN NOTE:
Example text for cautionary design notes.

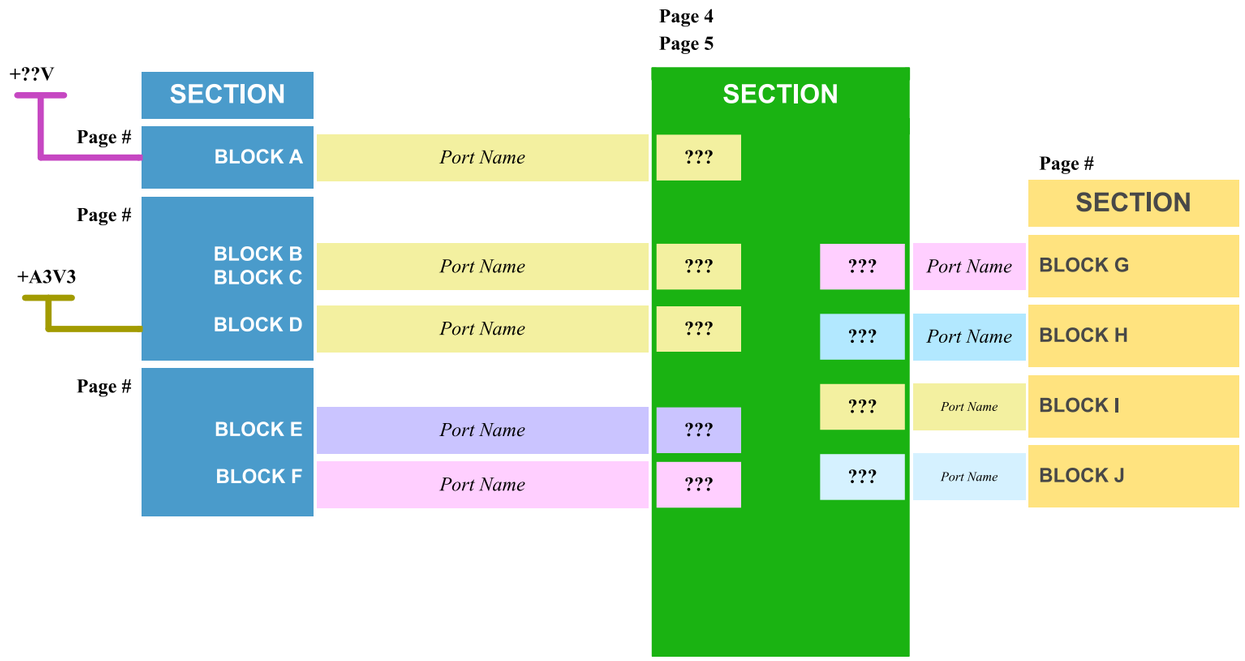
DESIGN NOTE:
Example text for critical design notes.

LAYOUT NOTE:
Example text for critical layout guidelines.

Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 44af293
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title:	File Name: Data-Acquisition-Board.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 22

[2] Block Diagram



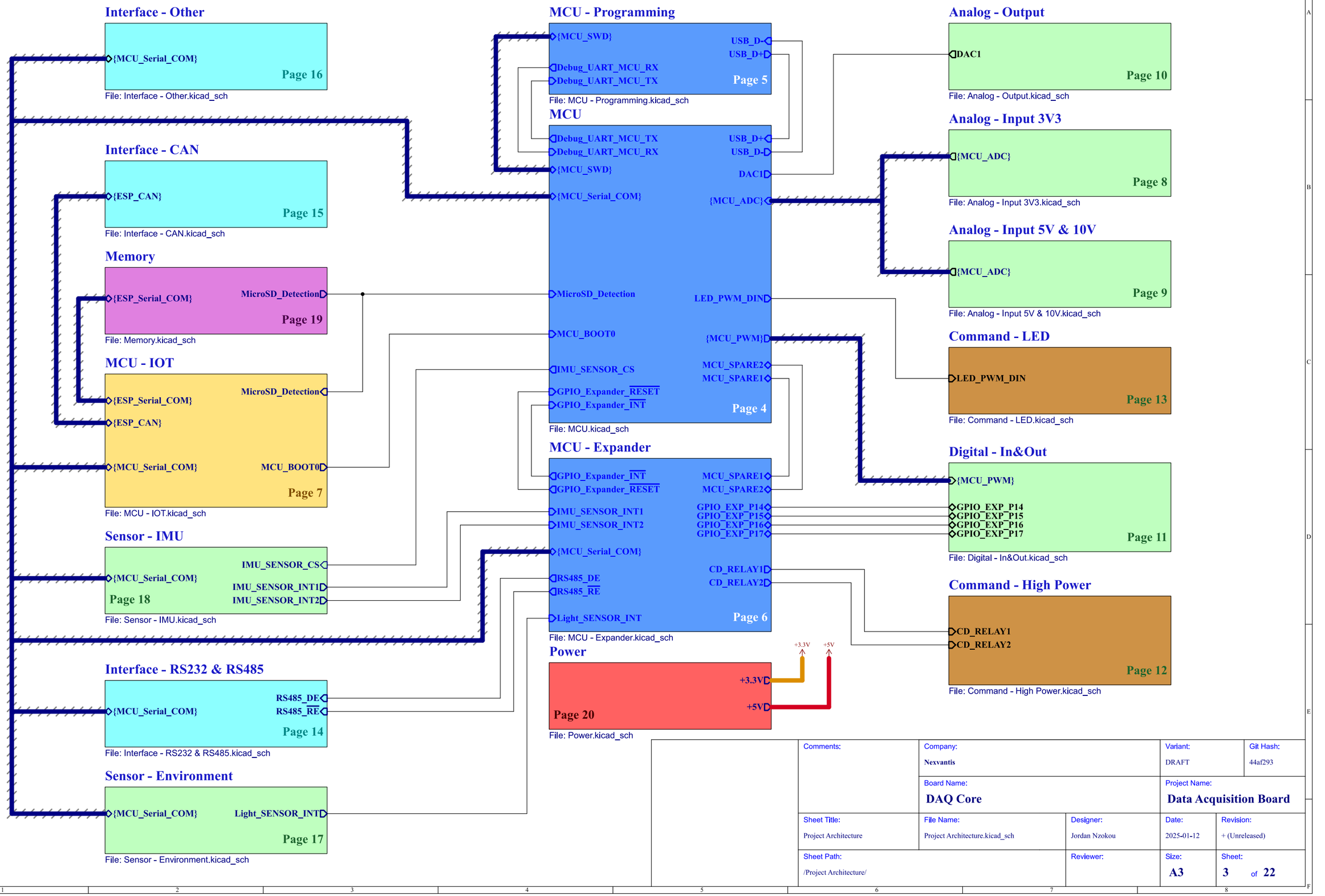


Target specifications:

Input voltage:	7V - 17V
Input power:	7400mW
Spec 3	??
Spec 4	??

Comments:	Company: **Nexvantis**		Variant: DRAFT	Git Hash: 44af293
	Board Name: **DAQ Core**		Project Name: **Data Acquisition Board**	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Block Diagram/		Reviewer:	Size: **A3**	Sheet: **2** of **22**

[3] Project Architecture

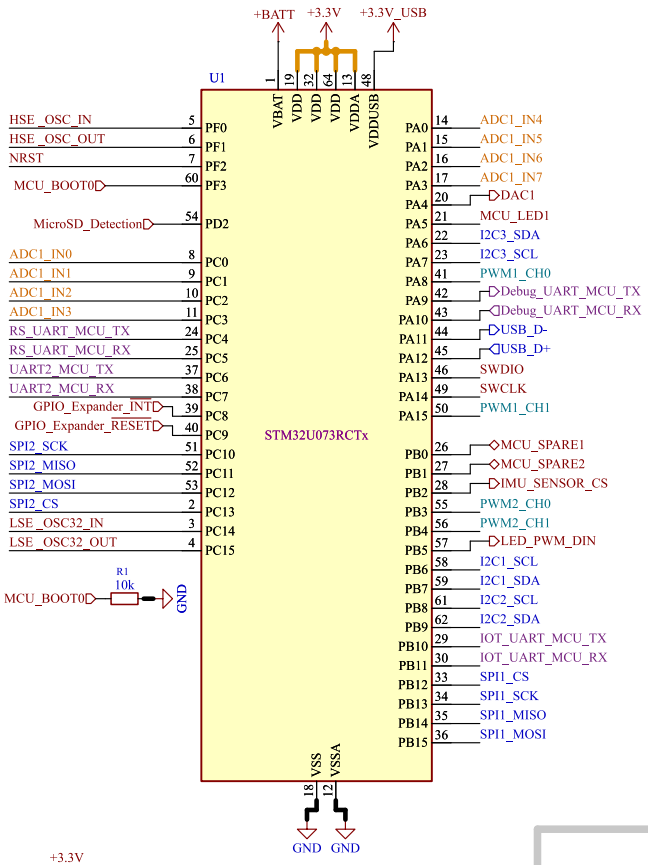


Comments:	Company: Nexvantis	Variant: DRAFT	Git Hash: 44af293
	Board Name: DAQ Core	Project Name: Data Acquisition Board	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12 / Revision: + (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: A3 / Sheet: 3 of 22

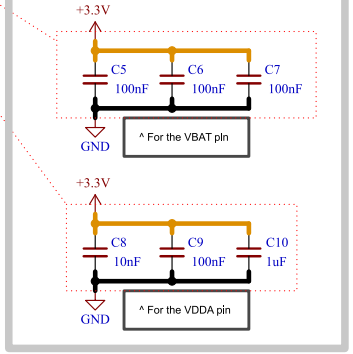
[4] MCU

LAYOUT NOTE:
Place each capacitor as close as to the relevant pin.

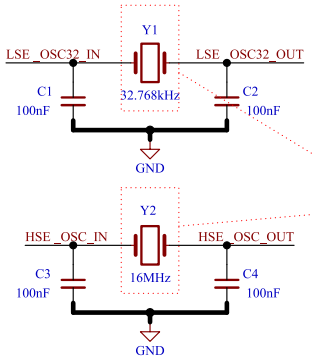
STM32U073RCTx



MCU Capacitor

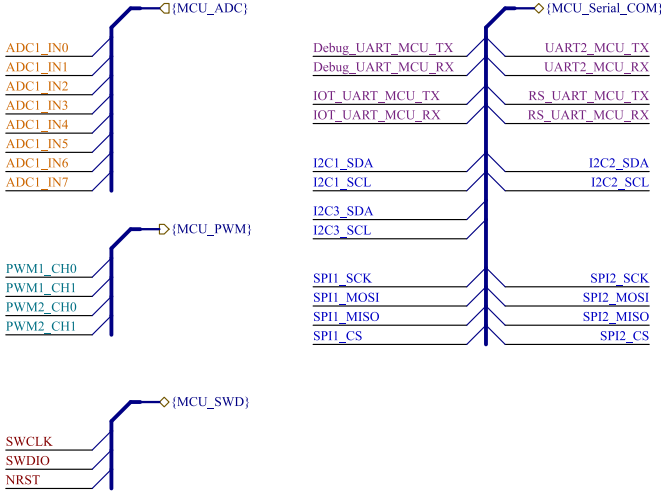


Quartz

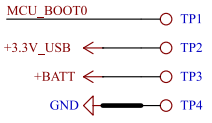


The values of the quartz crystals must be chosen with the utmost care, depending on the required speed of the microcontroller.

BUS

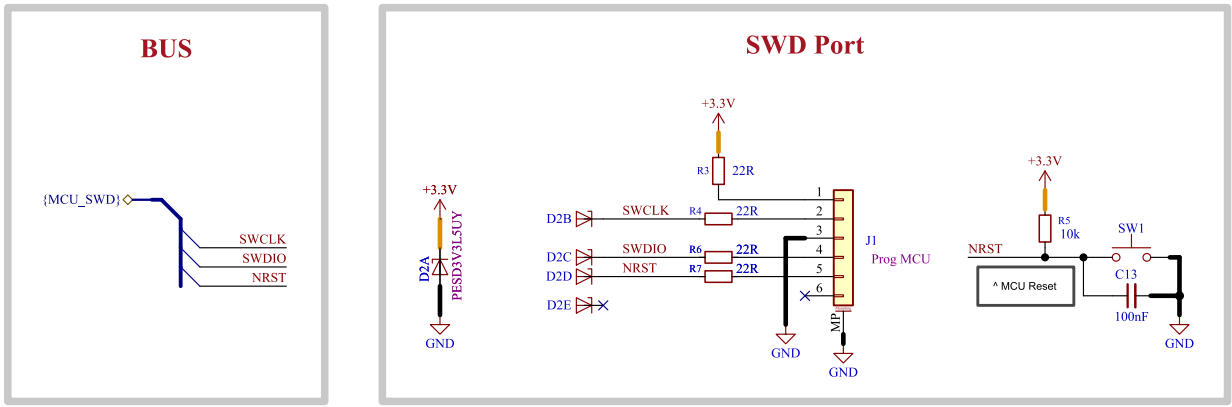


Test Points

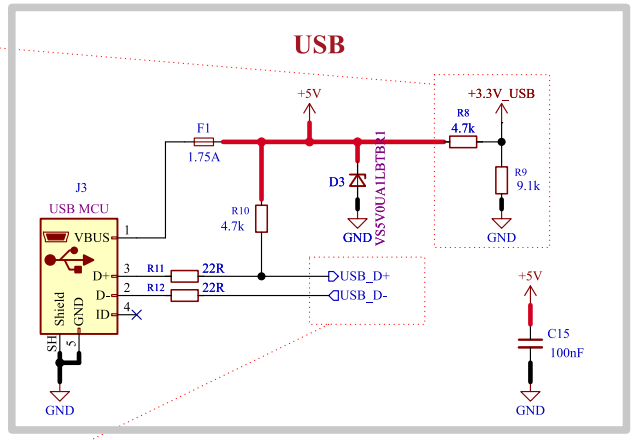


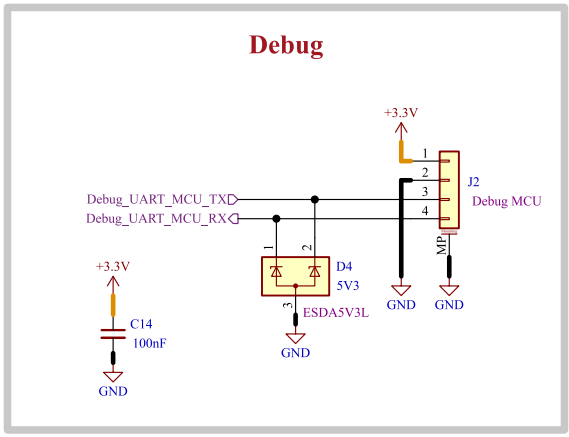
Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 44af293
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: MCU	File Name: MCU.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/MCU/		Reviewer:	Size: A4	Sheet: 4 of 22

[5] MCU - Programming



Up to now, the 3V3_USB voltage has been generated by a voltage divider powered by the 5 V USB supply.
Only the microcontroller is powered by the 3V3_USB voltage.



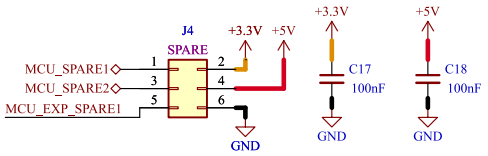


LAYOUT NOTE:
Route USB wire traces as a differential pair. Route away from analog signals.

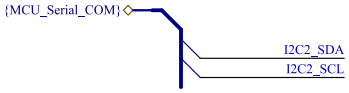
Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 44af293
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: MCU - Programming	File Name: MCU - Programming.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/MCU - Programming/		Reviewer:	Size: A4	Sheet: 5 of 22

[6] MCU - Expander

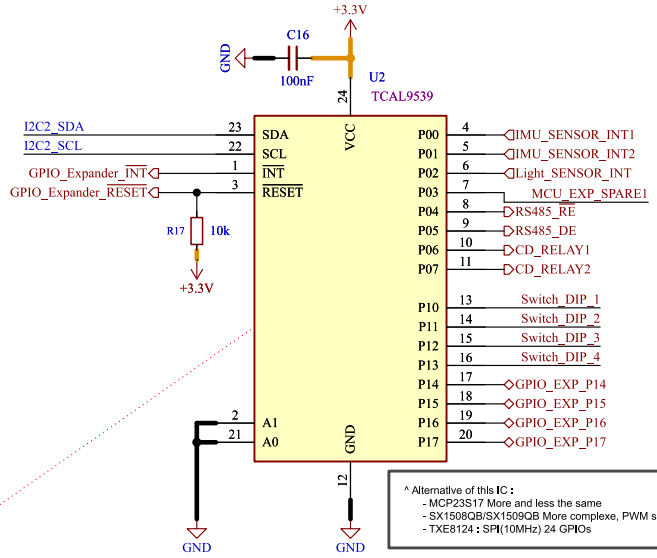
MCU SPARE CONNECTOR



BUS



GPIO Expander

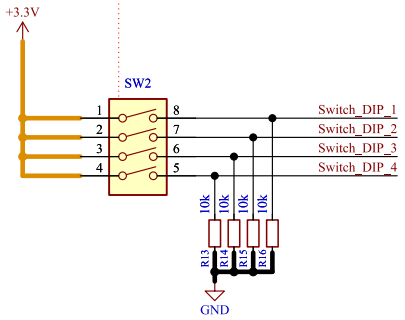


^ Alternative of this IC :
- MCP23S17 More and less the same
- SX1508QB/SX1509QB More complexe, PWM supported
- TXE8124 : SPI(10MHz) 24 GPIOs

The response time - that is, the delay between the expander receiving a trigger signal and the microcontroller receiving that signal - can be problematic if you require a rapid response.
This delay can be reduced by using an SPI expander in high-speed mode. This option will be included in a new version of the board.

This switch must be used to set different parameters in the software, so that the board can operate differently as required. Example: bypass the sensor on the board; enable/disable ultra-low-power mode.

Switch DIP



Test Points

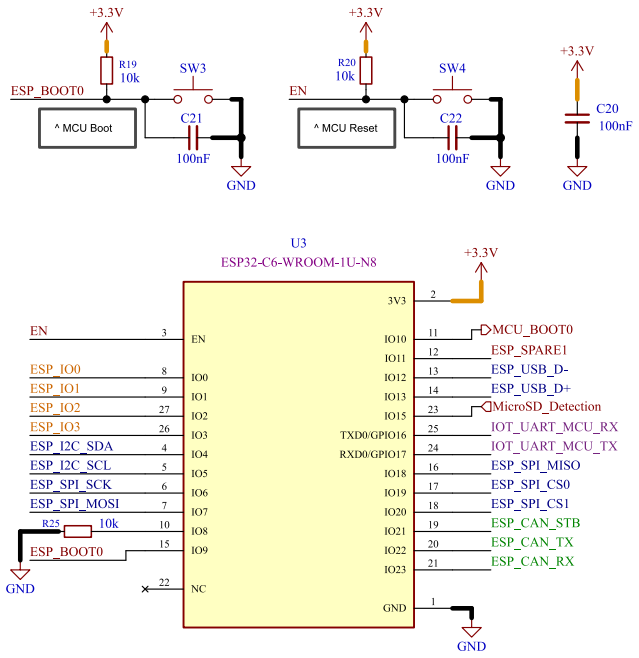


Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 44af293
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: MCU - Expander	File Name: MCU - Expander.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/MCU - Expander/		Reviewer:	Size: A4	Sheet: 6 of 22

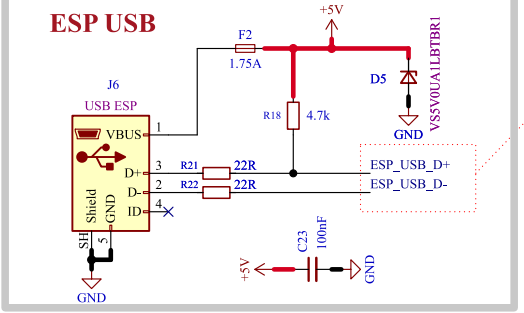
[7] MCU - IOT

LAYOUT NOTE:
Route USB wire traces as a differential pair. Route away from analog signals.

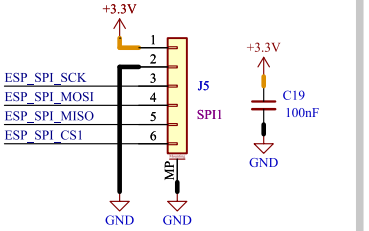
ESP32-C6-WROOM-1U



ESP USB



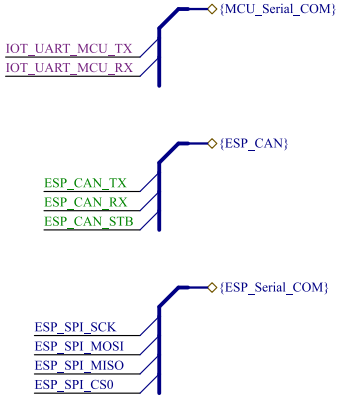
ESP SPI



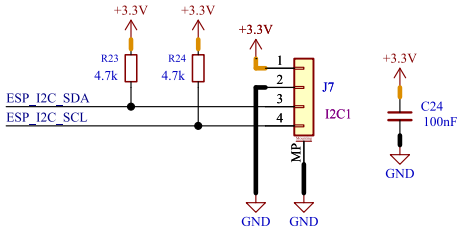
Test Points



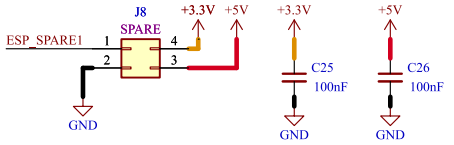
BUS



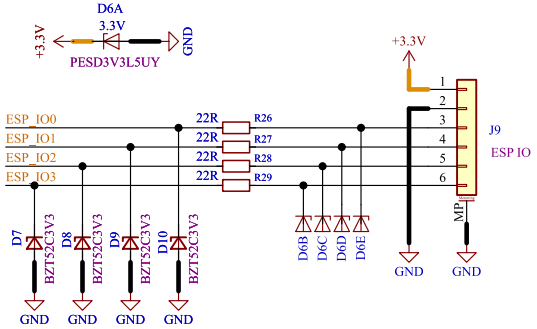
ESP I2C



ESP SPARE CONNECTOR



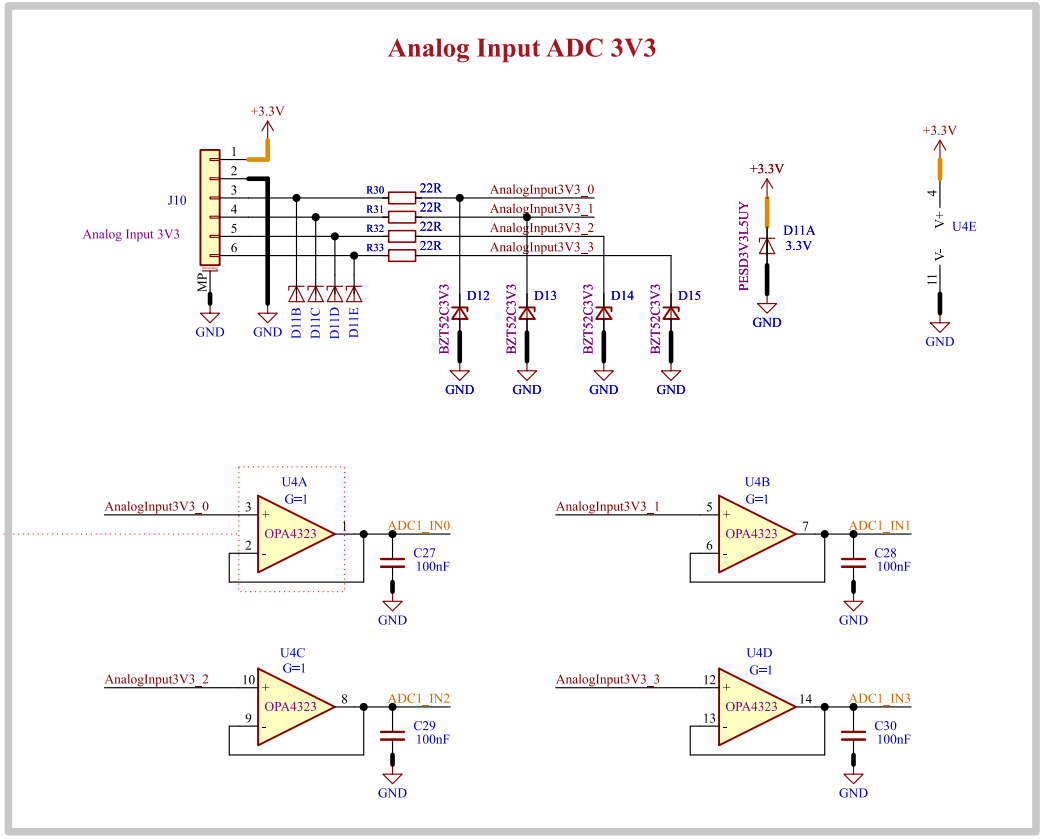
ESP IO



Comments: ESP32-C6-WROOM-1U Datasheet; TJA1042BT Datasheet	Company: Nexvantis		Variant: DRAFT	Git Hash: 44af293
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: MCU - IOT	File Name: MCU - IOT.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/MCU - IOT/		Reviewer:	Size: A4	Sheet: 7 of 22

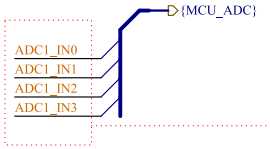
[8] Analog - Input 3V3

Analog Input ADC 3V3



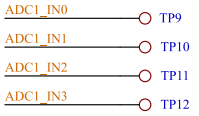
Addition of a voltage-follower operational amplifier to avoid an impedance matching problem.

BUS



LAYOUT NOTE:
Route analog traces away from digital signals.

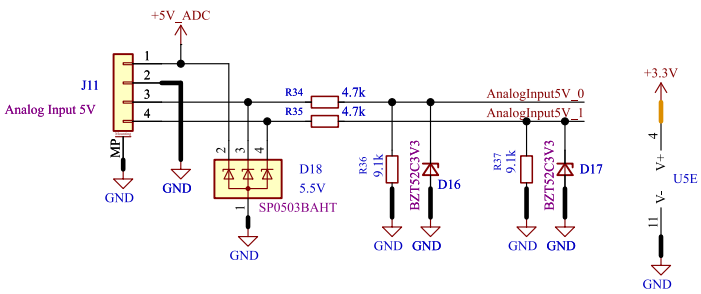
Test Points



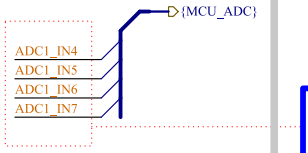
Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 44af293
	Board Name: **DAQ Core**		Project Name: **Data Acquisition Board**	
Sheet Title: Analog - Input 3V3	File Name: Analog - Input 3V3.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Analog - Input 3V3/		Reviewer:	Size: **A4**	Sheet: **8** of **22**

[9] Analog - Input 5V & 10V

Analog Input 5V

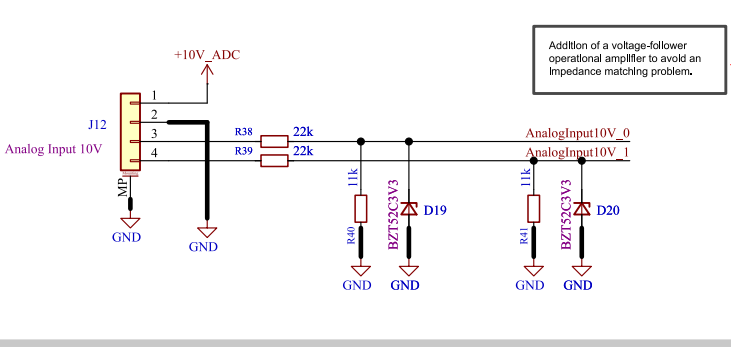


BUS

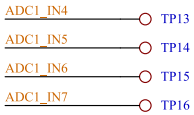


LAYOUT NOTE:
Route analog traces away from digital signals.

Analog Input 10V

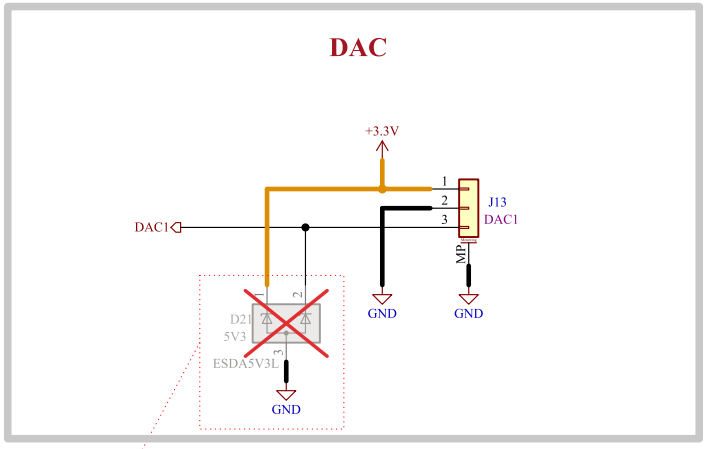


Test Points



Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 44af293
	Board Name: **DAQ Core**		Project Name: **Data Acquisition Board**	
Sheet Title: Analog - Input 5V & 10V	File Name: Analog - Input 5V & 10V.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Analog - Input 5V & 10V/		Reviewer:	Size: **A4**	Sheet: **9** of **22**

[10] Analog - Output

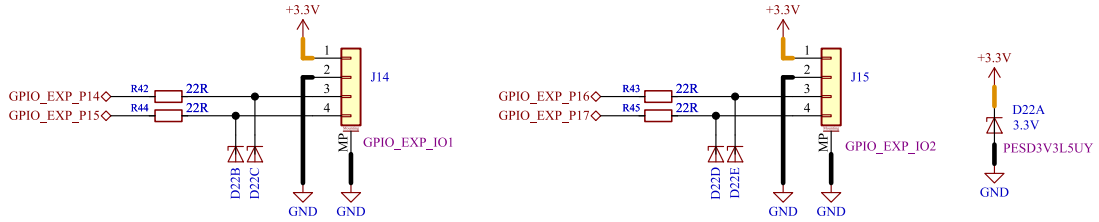


Adding an ESD diode may interfere with the DAC signal; that is why I have not included it by default.
Tests will determine whether or not it should be included.

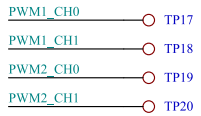
	Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 44af293
		Board Name: **DAQ Core**		Project Name: **Data Acquisition Board**	
	Sheet Title: Analog - Output	File Name: Analog - Output.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Project Architecture/Analog - Output/		Reviewer:	Size: **A4**	Sheet: **10** of **22**

[11] Digital - In&Out

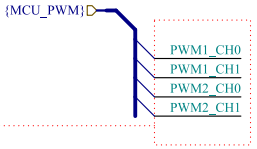
Digital I/O



Test Points

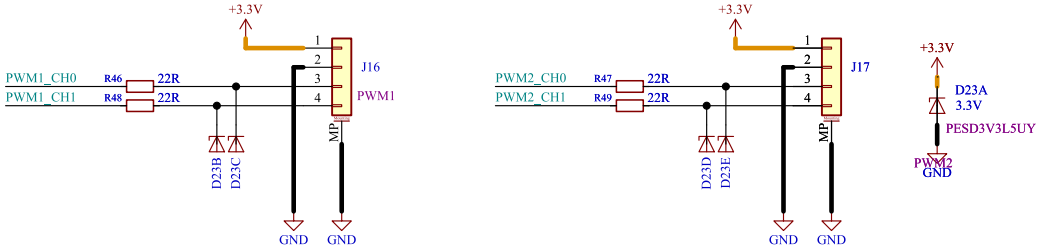


BUS



LAYOUT NOTE:
Route PWM traces away from analog signals.

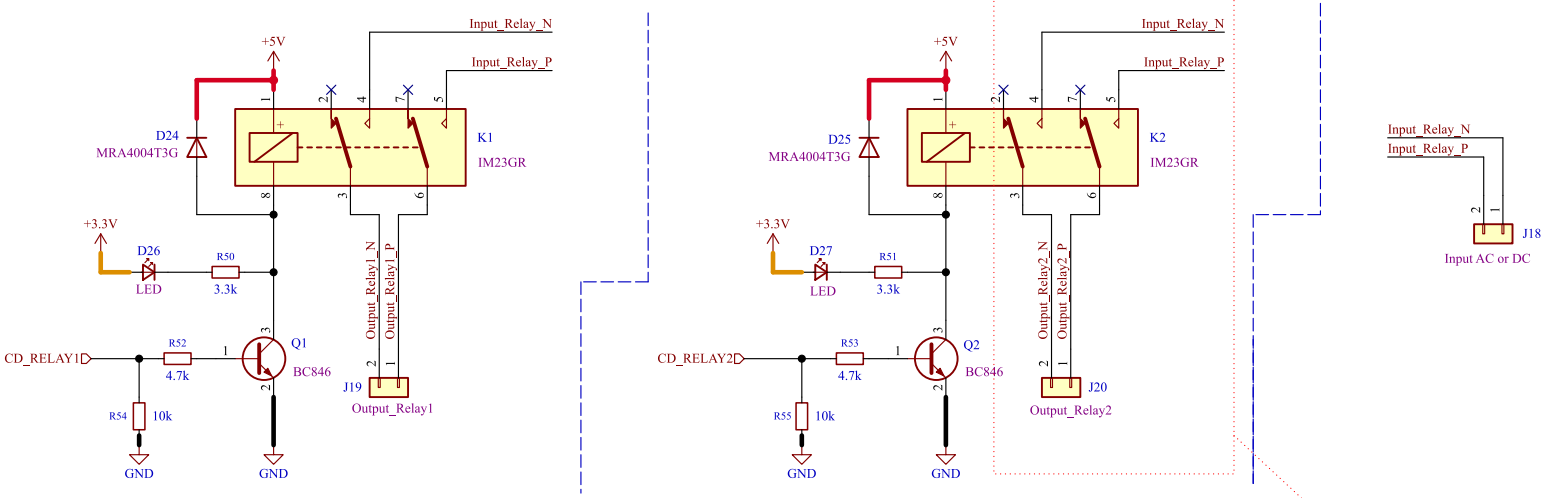
PWM



Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 44af293
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Digital - In&Out	File Name: Digital - In&Out.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Digital - In&Out/		Reviewer:	Size: A4	Sheet: 11 of 22

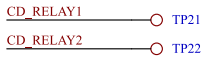
[12] Command - High Power

Relay



DESIGN NOTE:
Relays must be routed such that copper pours are Isolated from other copper areas.

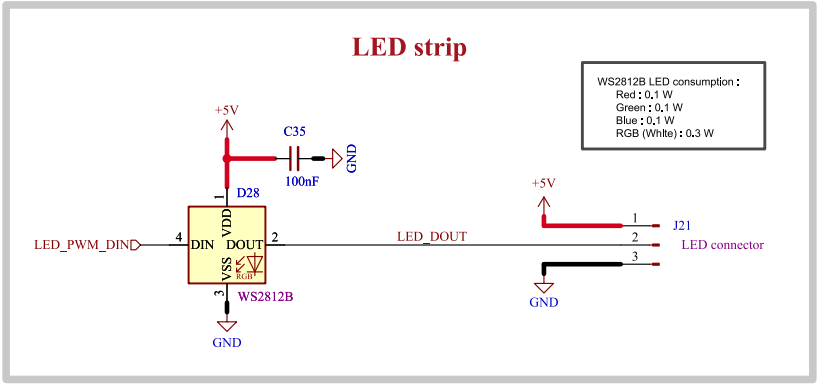
Test Points



Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 44af293
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Command - High Power	File Name: Command - High Power.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Command - High Power/		Reviewer:	Size: A4	Sheet: 12 of 22

[13] Command - LED

LED strip



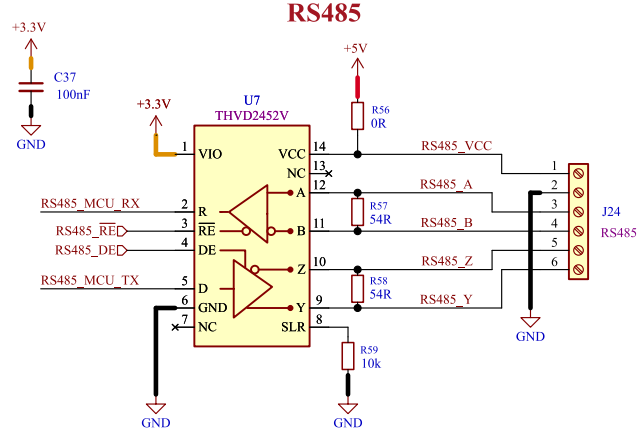
Test Points



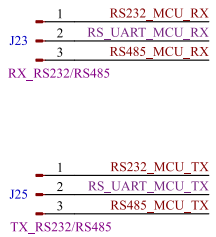
Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 44af293
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Command - LED	File Name: Command - LED.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Command - LED/		Reviewer:	Size: A4	Sheet: 13 of 22

[14] Interface - RS232 & RS485

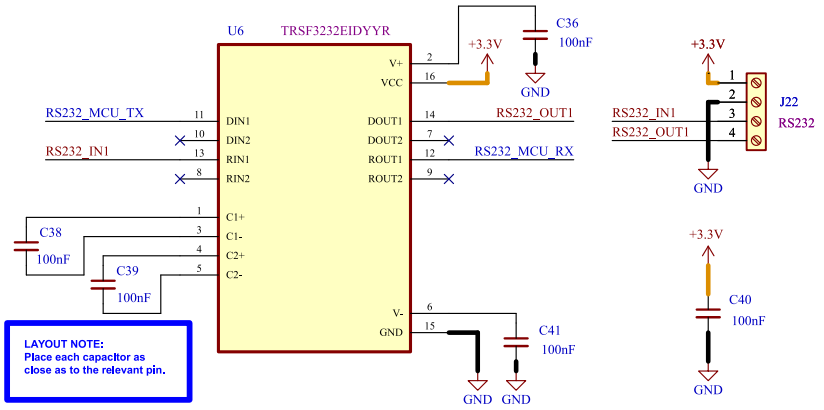
RS485



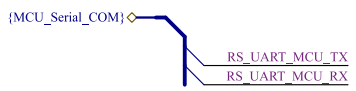
Selection RS232 or RS485



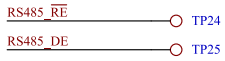
RS232



BUS



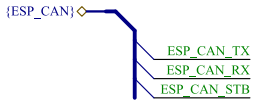
Test Points



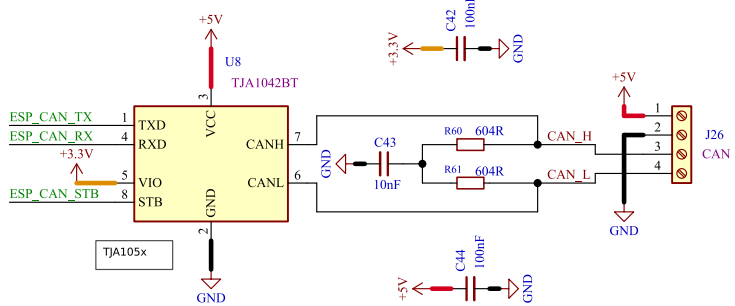
Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 44af293
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Interface - RS232 & RS485	File Name: Interface - RS232 & RS485.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Interface - RS232 & RS485/		Reviewer:	Size: A4	Sheet: 14 of 22

[15] Interface - CAN

BUS



ESP CAN



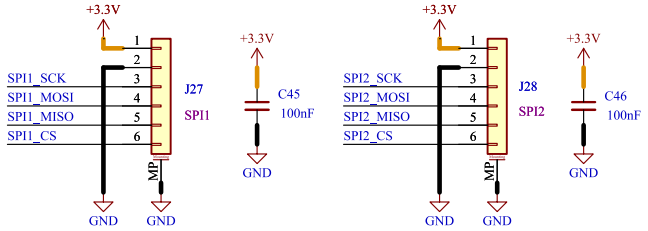
LAYOUT NOTE:
Route CAN_H and CAN_L traces as a differential pair. Avoid exposing them to signal that could cause interference.

Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 44af293
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Interface - CAN	File Name: Interface - CAN.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Interface - CAN/		Reviewer:	Size: A4	Sheet: 15 of 22

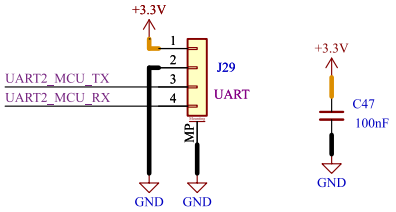
[16] Interface - Other

SPI1 must be used for devices that need high-speed communication.

SPI

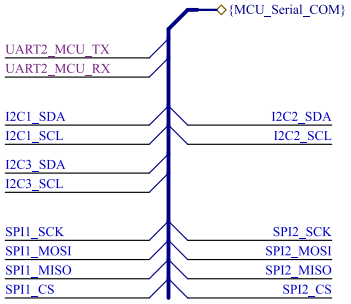


UART

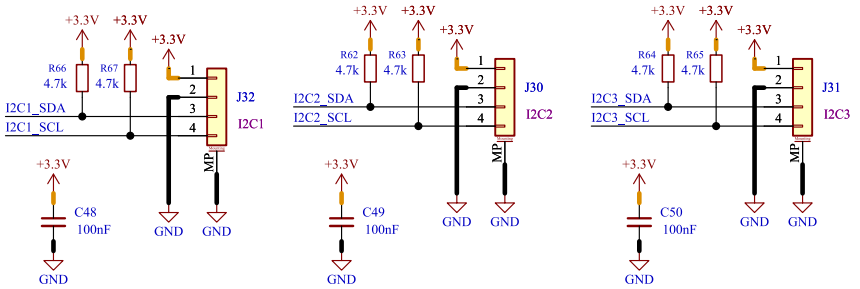


I2C1 must be used for devices that need high-speed communication.

BUS



I2C



Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 44af293
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Interface - Other	File Name: Interface - Other.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Interface - Other/		Reviewer:	Size: A4	Sheet: 16 of 22

[17] Sensor - Environment

BUS

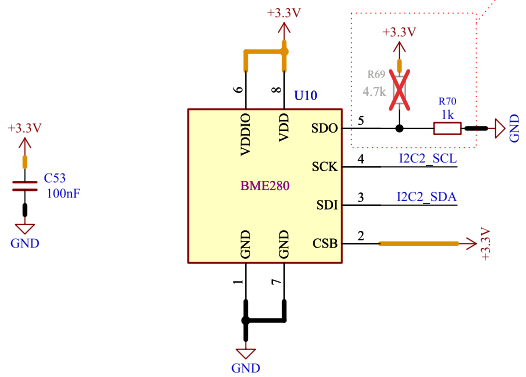


Test Points

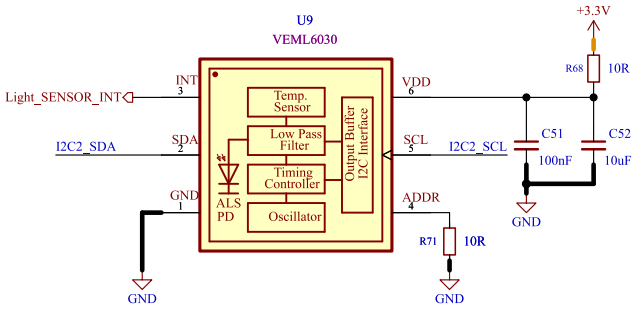


BME280 : Temperature, humidity and Pressure sensor

DESIGN NOTE:
Pin SDO is used to change the I2C adresse of the device :
High (VDDIO): 1110111 (0x77)
Low : 1110110 (0x76)



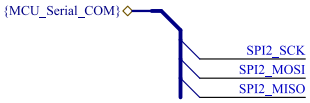
VEML6030 : Light sensor



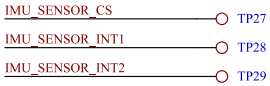
Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 44af293
	Board Name: **DAQ Core**		Project Name: **Data Acquisition Board**	
Sheet Title: Sensor - Environment	File Name: Sensor - Environment.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Sensor - Environment/		Reviewer:	Size: **A4**	Sheet: **17** of **22**

[18] Sensor - IMU

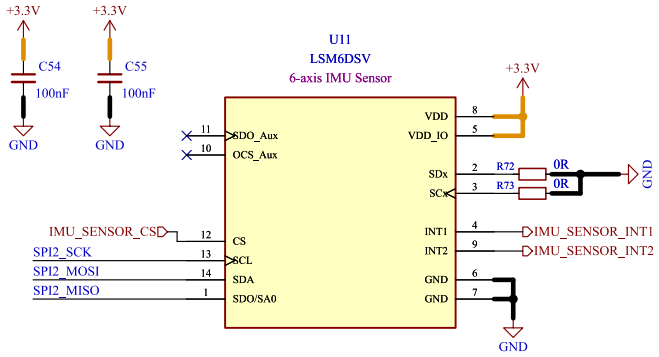
BUS



Test Points



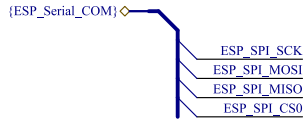
BME280 : IMU Sensor (Accelerometer and Gyroscope)



	Comments:	Company: Nexvantis	Variant: DRAFT	Git Hash: 44af293
		Board Name: DAQ Core	Project Name: Data Acquisition Board	
	Sheet Title: Sensor - IMU	File Name: Sensor - IMU.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12 / Revision: + (Unreleased)
	Sheet Path: /Project Architecture/Sensor - IMU/		Reviewer:	Size: A4 / Sheet: 18 of 22

[19] Memory

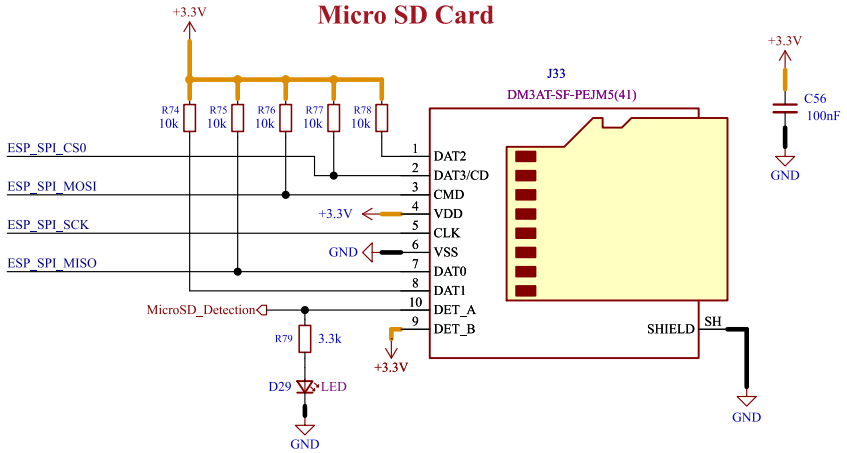
BUS



Test Points



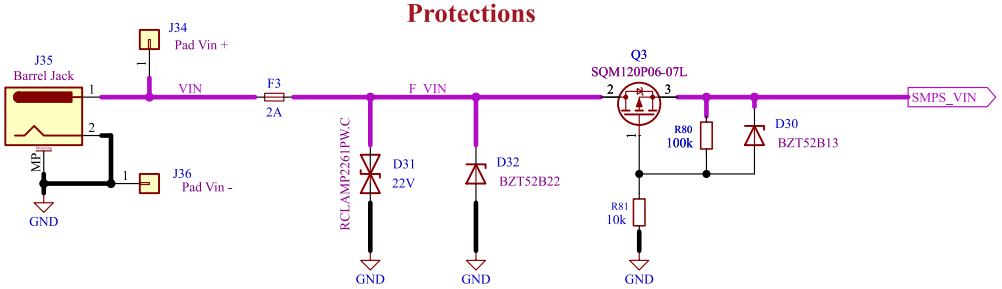
Micro SD Card



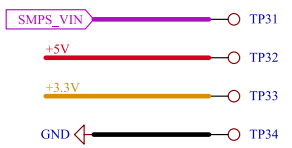
	Comments:	Company: Nexvantis	Variant: DRAFT	Git Hash: 44af293
		Board Name: DAQ Core	Project Name: Data Acquisition Board	
	Sheet Title: Memory	File Name: Memory.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12 / Revision: + (Unreleased)
	Sheet Path: /Project Architecture/Memory/		Reviewer:	Size: A4 / Sheet: 19 of 22

[20] Power

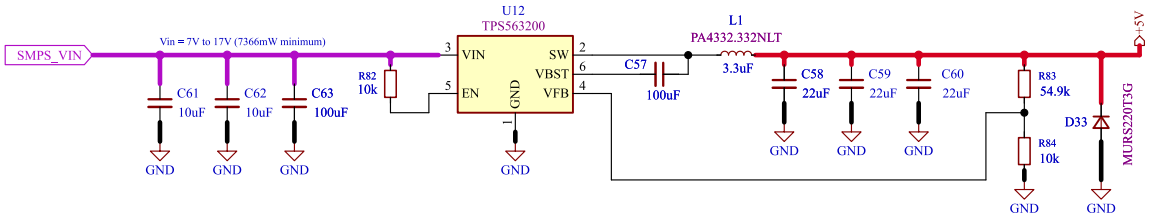
Protections



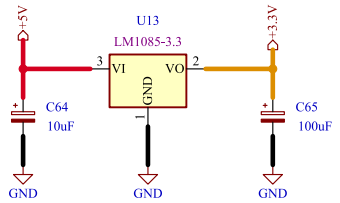
Test Points



5V SMPS Regulator



3V3 LDO Regulator



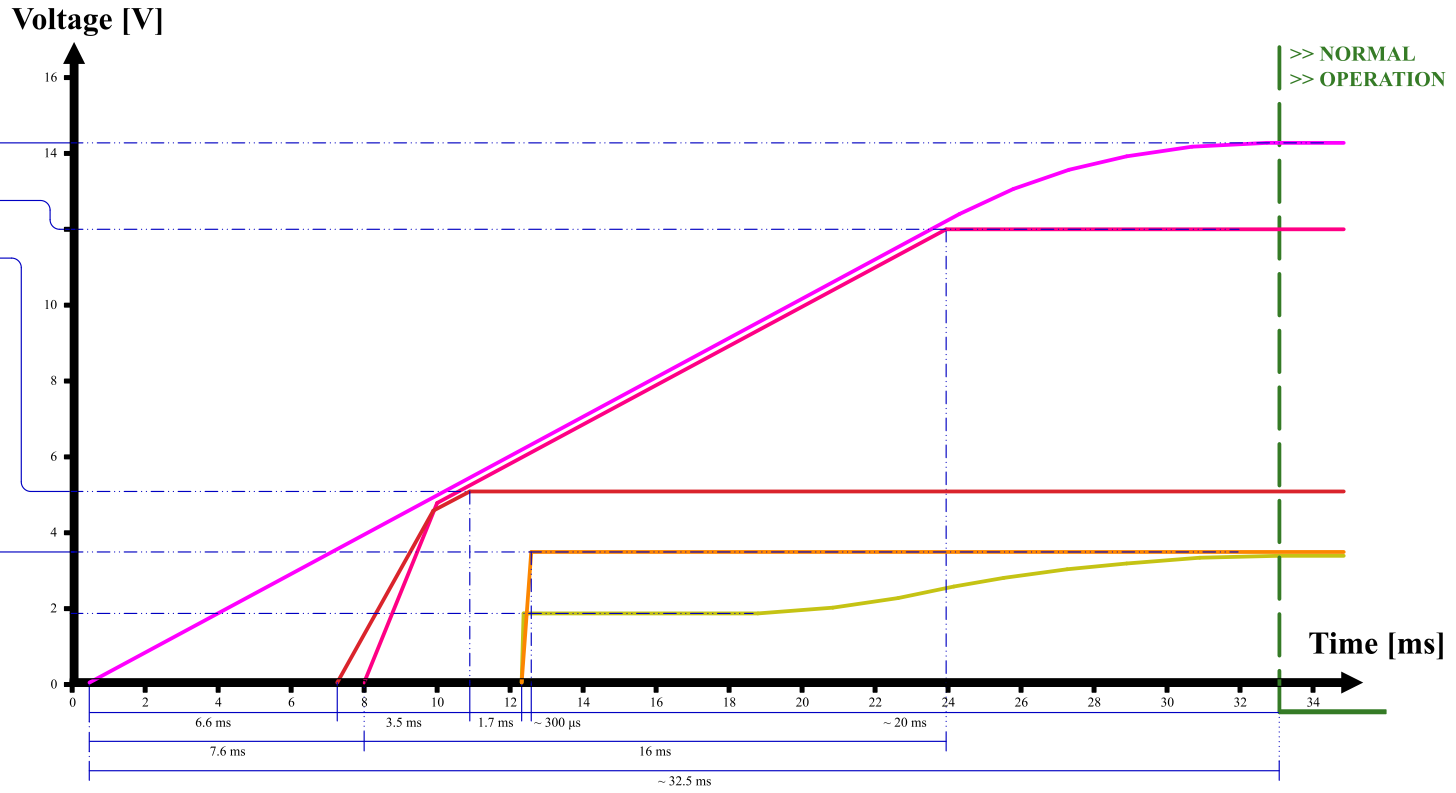
Protections:

->Overvoltage
->Voltage spike
->Overcurrent
->Reverse Polarity

Comments: LM1085-3.3 Datasheet; TPS563200 Datasheet	Company: Nexvantis		Variant: DRAFT	Git Hash: 44af293
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Power	File Name: Power.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Power/		Reviewer:	Size: A4	Sheet: 20 of 22

[21] Power - Sequencing

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%



Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 44af293
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 21 of 22

[22] Revision History

Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 44af293
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 22 of 22